

Multiple Choice

1. **Answer:** A

Options: A) $(x - 2)(x + 2)(x - 1)(x + 1)$; B) $(x+1)^4$; C) $(x-1)(x+1)^3$; D) $(x-1)^3(x+1)$

Note: Correct option: A - $(x - 2)(x + 2)(x - 1)(x + 1)$

2. **Answer:** B

Options: A) 4; B) 6; C) 8; D) 9

Note: Correct option: B - 6

3. **Answer:** C

Options: A) 1; B) 2; C) 3; D) 4

Note: Correct option: C - 3

4. **Answer:** C

Options: A) $1/2$; B) $7/12$; C) $5/8$; D) $2/3$

Note: Correct option: C - $5/8$

5. **Answer:** B

Options: A) (0, 0, 1); B) $(3/7, 3/14, 9/14)$; C) $(7/15, 8/15, 1/15)$; D) $(5/6, 1/3, 1/3)$

Note: Correct option: B - $(3/7, 3/14, 9/14)$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $4\mathbb{Z}, 2 + 4\mathbb{Z}$ '.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** The triangle is shown below: `[asy] pair A,B,C; A = (0,0); B = (5,0); C = (5,10); draw(A--B--C--A); draw(rightanglemark(C,B,A,16)); label("A",A,SW); label("B",B,SE); label("C",C,N); [/asy]` We have $\sin A = \frac{BC}{AC}$ and $\cos A = \frac{AB}{AC}$, so $\sin A = 2 \cos A$ gives us $\frac{BC}{AC} = 2 \cdot \frac{AB}{AC}$. Multiplying both sides by AC gives $BC = 2AB$, so $\frac{BC}{AB} = 2$. Finally, we have $\tan A = \frac{BC}{AB} = 2$. We also could have noted that $\tan A = \frac{\sin A}{\cos A} = \frac{2 \cos A}{\cos A} = 2$.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** I, II, and III. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.